



NeuroSpace

Complete evaluation export

Job ID	2fca297f-4fcc-4c49-bd06-4300089cc03d
Status	completed
Result timestamp	2026-05-11T18:16:56.970511
PDF generated	2026-05-11 20:41:13

Pipeline Configuration

Parameter	Value
model_types	rnn, qrnn
comparison_mode	pca
feature_selector	pca
qrnn_backend	pennylane
noise_enabled	False
noise_level	0.0000
mitigation_enabled	False
mitigation_runs	1
manual_overrides.rnn_epochs	20
manual_overrides.qrnn_epochs	10
manual_overrides.rnn_batch_size	16
manual_overrides.qrnn_batch_size	16

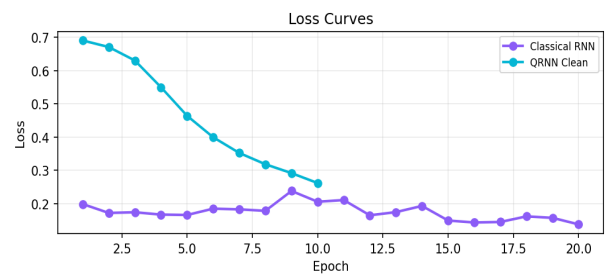
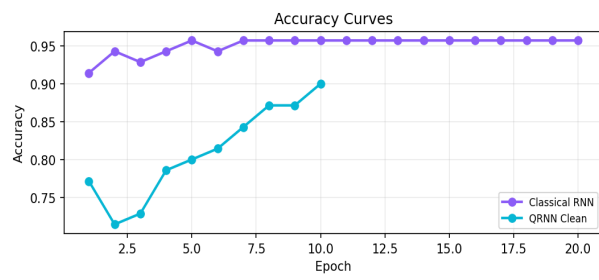
Best Model

Winner	RNN Accuracy	QRNN Accuracy	Improvement
RNN	93.57%	91.43%	2.14%

Evaluation Summary

Model	Accuracy	Precision	Recall	F1 Score	Loss
Classical RNN	93.57%	93.55%	93.57%	93.55%	0.1379
QRNN Clean	91.43%	91.99%	91.43%	91.54%	0.2620

Learning Curves



Classical RNN

Metric	Value
Accuracy	93.57%
Precision	93.55%
Recall	93.57%
F1 Score	93.55%
Loss	0.1379
Train Loss	0.0721
Val Loss	0.1379
Train Accuracy	97.55%
Val Accuracy	95.71%
Train Size	489
Val Size	70
Test Size	140

Classical RNN Confusion Matrix

Actual / Predicted	Class 0	Class 1
Class 0	88	4
Class 1	5	43

Classical RNN Classification Report

Class	Precision	Recall	F1 Score	Support
0	94.62%	95.65%	95.14%	92.0000
1	91.49%	89.58%	90.53%	48.0000
macro avg	93.06%	92.62%	92.83%	140.0000
weighted avg	93.55%	93.57%	93.55%	140.0000
Accuracy			93.57%	

Classical RNN Epoch History

Epoch	Train Loss	Val Loss	Train Acc	Val Acc
1	0.2628	0.1990	91.41%	91.43%
2	0.1399	0.1723	96.11%	94.29%
3	0.0924	0.1743	97.96%	92.86%
4	0.1219	0.1673	96.32%	94.29%
5	0.1048	0.1662	96.73%	95.71%
6	0.0904	0.1851	97.55%	94.29%

Epoch	Train Loss	Val Loss	Train Acc	Val Acc
7	0.0907	0.1830	97.75%	95.71%
8	0.1054	0.1786	97.75%	95.71%
9	0.1134	0.2382	96.93%	95.71%
10	0.0972	0.2059	96.93%	95.71%
11	0.1250	0.2110	96.52%	95.71%
12	0.1061	0.1652	97.14%	95.71%
13	0.0664	0.1748	97.96%	95.71%
14	0.0676	0.1933	97.96%	95.71%
15	0.0601	0.1495	97.75%	95.71%
16	0.0771	0.1436	97.34%	95.71%
17	0.0595	0.1451	98.16%	95.71%
18	0.0729	0.1619	97.96%	95.71%
19	0.0825	0.1575	97.14%	95.71%
20	0.0721	0.1379	97.55%	95.71%

QRNN Clean

Metric	Value
Accuracy	91.43%
Precision	91.99%
Recall	91.43%
F1 Score	91.54%
Loss	0.2620
Train Loss	0.2173
Val Loss	0.2620
Train Accuracy	94.48%
Val Accuracy	90.00%
Train Size	489
Val Size	70
Test Size	140

QRNN Clean Confusion Matrix

Actual / Predicted	Class 0	Class 1
Class 0	83	9
Class 1	3	45

QRNN Clean Classification Report

Class	Precision	Recall	F1 Score	Support
0	96.51%	90.22%	93.26%	92.0000
1	83.33%	93.75%	88.24%	48.0000
macro avg	89.92%	91.98%	90.75%	140.0000
weighted avg	91.99%	91.43%	91.54%	140.0000
Accuracy			91.43%	

QRNN Clean Epoch History

Epoch	Train Loss	Val Loss	Train Acc	Val Acc
1	0.7054	0.6900	49.28%	77.14%
2	0.6776	0.6703	74.64%	71.43%
3	0.6401	0.6299	76.69%	72.86%
4	0.5667	0.5496	82.41%	78.57%
5	0.4707	0.4639	85.48%	80.00%
6	0.3930	0.3994	87.93%	81.43%

Epoch	Train Loss	Val Loss	Train Acc	Val Acc
7	0.3388	0.3523	91.00%	84.29%
8	0.2949	0.3184	92.23%	87.14%
9	0.2555	0.2918	93.25%	87.14%
10	0.2173	0.2620	94.48%	90.00%